

Biexciton Binding and Radiative Lifetimes in Strongly Oblate Semi-Ellipsoidal Quantum Dots

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Abstract

We investigate the ground exciton and biexciton states in a hard-wall, strongly oblate semi-ellipsoidal GaAs quantum dot. Their energies are evaluated variationally using correlated trial states constructed from analytical single-particle wave functions, together with a ground-orbital-matched importance transformation and adaptive quasi-Monte Carlo integration. A noise-aware coordinate pattern search with two-budget verification is used to optimize the variational parameters along intersecting axial and lateral geometry scans. Because the effective-mass hard-wall model is intended to isolate geometrical effects, the principal results are qualitative trends rather than absolute predictions for a particular fabricated dot. The axial dimension controls the absolute confinement energy much more strongly than the lateral dimension, whereas enlargement along either direction weakens the biexciton binding. The central binding estimate remains positive throughout the sampled range but is not resolved from zero at the widest lateral geometry. Increasing either semiaxis enhances the exciton coincidence overlap and oscillator strength and therefore shortens the calculated radiative lifetimes. The lateral dependence remains pronounced where the axial dependence approaches saturation. These results distinguish the dominant axial control of the confinement-energy scale from the important lateral control of weak biexciton binding and radiative strength.

Keywords: semi-ellipsoidal quantum dot; exciton; biexciton; binding energy; variational method; quasi-Monte Carlo integration; radiative lifetime

1 Introduction

Semiconductor quantum dots combine discrete, atom-like carrier spectra with optical and electronic properties that can be tailored through composition, size, and shape. This tunability supports applications in displays and lighting, lasers, sensing, communications, energy conversion, bioimaging, and quantum information [1]. In quantum-photonic applications, individual dots can act as sources of single photons and photon pairs [2]. The radiative biexciton–exciton cascade produces time-correlated photon pairs [3, 4] and can generate polarization-entangled photons [5]. The biexciton binding energy sets the spectral separation of the two cascade transitions, while the exciton and biexciton radiative lifetimes set their emission time scales and influence the attainable photon-generation rate.

The energy of a confined biexciton is not fixed by a single Coulomb scale. Electron–hole attraction competes with electron–electron and hole–hole repulsion, while the carrier masses and confinement geometry determine how strongly the four particles can rearrange. Consequently, the sign and magnitude of the binding energy can vary strongly with the material and dot structure [6–8]. This sensitivity makes the biexciton a useful probe of few-particle correlations, but it also makes its numerical evaluation demanding: the binding energy is a small difference between two much larger correlated energies.

Strongly oblate dots are especially suitable for separating the roles of axial and lateral confinement. Their two semiaxes provide independent geometrical controls, while the large separation between axial and in-plane energy scales permits an adiabatic treatment of the single-particle

states. Correlated few-particle states and radiative lifetimes have been studied in strongly oblate full ellipsoids [9], including their nonlinear optical response [10]. For semi-ellipsoidal dots, previous calculations addressed electronic and excitonic states, binding and recombination energies [11], single-particle states and interband absorption [12], and, more recently, perturbative exciton binding, absorption, and photoluminescence in a hard-wall lens-shaped model [13]. These semi-ellipsoidal studies do not treat the correlated biexciton or its associated radiative lifetime.

The purpose of this work is to determine how the lateral semiaxis a and the axial semiaxis c control the correlated exciton and biexciton energies, biexciton binding, exciton coincidence overlap, and radiative lifetimes in a strongly oblate semi-ellipsoidal GaAs quantum dot. We restrict the calculation to the single-particle ground configuration and evaluate correlated variational states using a state-matched importance transformation, adaptive quasi-Monte Carlo integration, and a two-phase variational search. Two intersecting geometry scans vary one semiaxis at a time to distinguish axial and lateral trends. The idealized hard-wall effective-mass model is intended to isolate those geometrical effects rather than reproduce a particular fabricated dot. Accordingly, the principal results are the qualitative dependences on dot geometry; the absolute numerical values should be interpreted within the assumptions of the model. Particular attention is paid to whether the small biexciton binding remains resolved at the available numerical precision.

2 Model and theoretical framework

The interacting exciton and biexciton ground-state energies are obtained with the variational method: trial correlation factors are optimized by minimizing the corresponding Hamiltonian expectation values. We first establish the single-particle confinement states on which these trial states are built.

2.1 Geometry, units, and single-particle ground state

We consider a GaAs hard-wall semi-ellipsoidal quantum dot with cylindrical coordinates $\mathbf{r} = (r, \phi, z)$ and confinement potential

$$V_{\text{conf}}(r, z) = \begin{cases} 0, & r^2/a^2 + z^2/c^2 \leq 1 \text{ and } z \geq 0, \\ \infty, & \text{otherwise,} \end{cases} \quad c/a \ll 1. \quad (1)$$

The in-plane and axial semiaxes are a and c , respectively, as illustrated in Fig. 1.

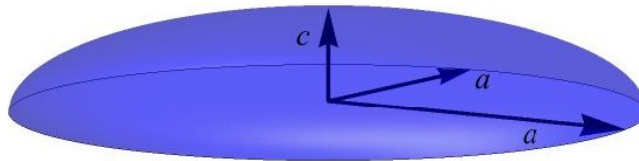


Figure 1: Semi-ellipsoidal quantum-dot geometry, with lateral semiaxis a and axial semiaxis c .

Lengths and energies are expressed in the effective electron Bohr radius and Rydberg,

$$r_B = \frac{4\pi\epsilon_0\epsilon_r\hbar^2}{m_e e^2}, \quad E_R = \frac{\hbar^2}{2m_e r_B^2}. \quad (2)$$

Once expressed in these units, the dimensionless energy problem depends on the material only through the effective-mass ratio $\eta = m_e/m_h$. The numerical GaAs calibration is introduced together with the radiative-lifetime parameters in section 2.4.

Because the axial confinement is much stronger than the in-plane confinement, the single-particle problem is treated adiabatically [12]. At fixed r , the available height is

$$z_{\text{max}}(r) = c\sqrt{1 - r^2/a^2}, \quad (3)$$

and the lowest axial factor is

$$\chi_1(z; r) = \sqrt{\frac{2}{z_{\max}(r)}} \sin \left[\frac{\pi z}{z_{\max}(r)} \right]. \quad (4)$$

For a carrier of mass m_q with $q = e, h$ the axial eigenenergy acts as the adiabatic potential for the slow planar motion,

$$U_q(r) = \frac{m_e}{m_q} \frac{\pi^2}{z_{\max}^2(r)} = \frac{m_e}{m_q} \frac{\pi^2}{c^2} \left(1 + \frac{r^2}{a^2} + O(r^4/a^4) \right), \quad (5)$$

where lengths and energies are in the effective units of Eq. (2). Thus, after subtracting the constant axial energy, the planar harmonic-oscillator Hamiltonian is

$$H_{q,\parallel} = \frac{m_e}{m_q} \left[-\nabla_{\parallel}^2 + \frac{\pi^2 r^2}{a^2 c^2} \right]. \quad (6)$$

Writing $\lambda = \pi/(ac)$, its normalized ground-state component and the corresponding axial and planar ground-state energies are

$$\varphi_{q,\parallel}(r) = \sqrt{\frac{\lambda}{\pi}} e^{-\lambda r^2/2}, \quad \varepsilon_{q,z} = \frac{m_e}{m_q} \frac{\pi^2}{c^2}, \quad \varepsilon_{q,\parallel} = \frac{m_e}{m_q} \frac{2\pi}{ac}. \quad (7)$$

The total ground-state confinement energy of carrier q is therefore

$$E_q = \varepsilon_{q,z} + \varepsilon_{q,\parallel} = \frac{m_e}{m_q} \left(\frac{\pi^2}{c^2} + \frac{2\pi}{ac} \right). \quad (8)$$

The normalized carrier ground orbital is therefore $\psi_q(r, \phi, z) = \chi_1(z; r) \varphi_{q,\parallel}(r)$. Thus the uncorrelated exciton and biexciton confinement energies are

$$E_X^{(0)} = E_e + E_h, \quad E_{XX}^{(0)} = 2E_e + 2E_h = 2E_X^{(0)}. \quad (9)$$

The band gap is not included in these energies.

2.2 Correlated exciton

We denote electron coordinates by $1, 2$ and hole coordinates by a, b ; the ground exciton uses coordinates 1 and a . Pair distances are the three-dimensional Euclidean distances $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$. In the effective units of Eq. (2), its Hamiltonian is

$$H_X = h_e(1) + h_h(a) - \frac{2}{r_{1a}}, \quad h_e \psi_e = E_e \psi_e, \quad h_h \psi_h = E_h \psi_h. \quad (10)$$

The uncorrelated state $\Psi_{0X} = \psi_e(1) \psi_h(a)$ is therefore an eigenstate of $h_e + h_h$ with energy $E_X^{(0)} = E_e + E_h$. Electron-hole correlation is introduced through

$$\Psi_X = \Psi_{0X} F_X, \quad F_X = \exp(-\alpha_X r_{1a}), \quad (11)$$

where $\alpha_X > 0$ is varied independently at each geometry. The variational energy is evaluated from

$$E_X = \frac{\langle \Psi_X | H_X | \Psi_X \rangle}{\langle \Psi_X | \Psi_X \rangle}. \quad (12)$$

The denominator is

$$N_X = \int |\Psi_{0X}|^2 F_X^2 d\tau = \int |\Psi_{0X}|^2 e^{-2\alpha_X r_{1a}} d\tau. \quad (13)$$

Using the single-particle eigenvalue equations in Eq. (10) and integrating the kinetic terms by parts gives

$$E_X = E_X^{(0)} + \frac{1}{N_X} \int |\Psi_{0X}|^2 \left[|\nabla_1 F_X|^2 + \frac{m_e}{m_h} |\nabla_a F_X|^2 - \frac{2F_X^2}{r_{1a}} \right] d\tau. \quad (14)$$

Since $|\nabla_1 r_{1a}| = |\nabla_a r_{1a}| = 1$, the kinetic contribution generated by F_X is

$$K_X(\alpha_X) = \alpha_X^2 \left(1 + \frac{m_e}{m_h} \right). \quad (15)$$

Defining the Coulomb numerator as

$$V_X = \int |\Psi_{0X}|^2 F_X^2 \left(-\frac{2}{r_{1a}} \right) d\tau \quad (16)$$

reduces Eq. (14) to

$$E_X = E_X^{(0)} + \Delta E_X, \quad \Delta E_X = K_X + \frac{V_X}{N_X}. \quad (17)$$

For each dot geometry, the remaining integrals N_X and V_X are evaluated numerically.

2.3 Correlated biexciton

The two-electron–two-hole interaction in effective units is

$$V_{XX} = 2 \left(\frac{1}{r_{12}} + \frac{1}{r_{ab}} - \frac{1}{r_{1a}} - \frac{1}{r_{1b}} - \frac{1}{r_{2a}} - \frac{1}{r_{2b}} \right). \quad (18)$$

For the biexciton we use the four-parameter ansatz employed by Takagahara [6],

$$\Psi_{XX} = \Psi_{0XX} F_{XX}, \quad \Psi_{0XX} = \psi_e(1)\psi_e(2)\psi_h(a)\psi_h(b), \quad (19)$$

with the exchange-symmetric correlation factor

$$\begin{aligned} F_{XX} &= G(P + Q), \\ G &= r_{ab}^\gamma e^{-\delta r_{ab}}, \\ P &= \exp[-\alpha(r_{1a} + r_{2b}) - \beta(r_{1b} + r_{2a})], \\ Q &= \exp[-\beta(r_{1a} + r_{2b}) - \alpha(r_{1b} + r_{2a})]. \end{aligned} \quad (20)$$

The four variational parameters $(\alpha, \beta, \gamma, \delta)$ control the direct and cross electron–hole pairings and the hole–hole correlation.

To write the kinetic numerator compactly, define $\mathbf{g}_i^P = \nabla_i \ln(GP)$ and $\mathbf{g}_i^Q = \nabla_i \ln(GQ)$. For electron coordinates the derivatives of G vanish, whereas for hole coordinates they contain the logarithmic derivative of G in Eq. (20). Unlike the exciton kinetic contribution, the biexciton kinetic numerator has no analytical form and is written as

$$K_{XX} = \int |\Psi_{0XX}|^2 G^2 \left[\sum_{i=1,2} |P\mathbf{g}_i^P + Q\mathbf{g}_i^Q|^2 + \frac{m_e}{m_h} \sum_{j=a,b} |P\mathbf{g}_j^P + Q\mathbf{g}_j^Q|^2 \right] d\tau. \quad (21)$$

The energy is then

$$E_{XX} = E_{XX}^{(0)} + \Delta E_{XX}, \quad \Delta E_{XX} = \frac{K_{XX} + \langle \Psi_{XX} | V_{XX} | \Psi_{XX} \rangle}{\langle \Psi_{XX} | \Psi_{XX} \rangle}. \quad (22)$$

In the numerical calculation, the kinetic contribution in Eq. (21) and the potential contribution are combined and evaluated in a single integral. No direct/cross or particle-exchange multiplicities are introduced in the final estimator.

We define the biexciton binding energy as

$$E_{\text{bind}} = 2E_X - E_{XX} = 2\Delta E_X - \Delta E_{XX}. \quad (23)$$

With this convention $E_{\text{bind}} > 0$ denotes a bound biexciton. The single-particle part cancels exactly, making Eq. (23) a small difference of correlation corrections.

2.4 Radiative lifetimes

Following Hayrapetyan *et al.* [9], we calculate the exciton and biexciton radiative lifetimes from

$$\tau_X = \frac{2\pi\epsilon_0 m_e c_0^3 \hbar^2}{\sqrt{\epsilon_r} e^2 E_{\text{exc}}^2 f_X}, \quad \tau_{XX} = \frac{\tau_X}{4}. \quad (24)$$

The relation $\tau_{XX} = \tau_X/4$ follows the biexciton decay-channel convention of Ref. [9]; it is not an independently optimized quantity.

Evaluation of τ_X requires the oscillator strength and transition energy, which are

$$f_X = \frac{E_P}{E_{\text{exc}}} |M_X|^2, \quad E_{\text{exc}} = E_g + E_X, \quad (25)$$

where E_g is the band gap and E_X is converted from effective Rydbergs to electronvolts using Eq. (2). The remaining quantity M_X is the electron-hole coincidence overlap. At coincidence the exciton Jastrow factor in Eq. (11) is unity, and because the electron and hole ground orbitals are matched in the present model, the normalized overlap reduces to

$$M_X = \frac{1}{\sqrt{N_X}}, \quad |M_X|^2 = \frac{1}{N_X}. \quad (26)$$

Thus the lifetime can be obtained from the already calculated exciton norm; no additional oscillator-strength integral is required.

For all numerical results we use the single GaAs parameter set $\epsilon_r = 12.8$, $m_e = 0.067m_0$, and $m_h = 0.45m_0$ [14], which gives $r_B \simeq 10.11$ nm and $E_R \simeq 5.564$ meV. The same ϵ_r is used for Coulomb screening and for the dielectric factor in Eq. (24). The radiative-lifetime calculation additionally uses $E_g = 1.519$ eV and $E_P = 22.71$ eV, following the optical model of Ref. [9].

3 Numerical method and validation

3.1 Geometry and integration parametrization

We study the seven geometries listed in Table 1. The common point (5, 1) links a scan of axial confinement to a scan of lateral confinement.

Table 1: Quantum-dot semiaxes used in the axial and lateral scans, in effective Bohr radii.

Scan	a	c
Axial	5	0.5, 1, 1.5, 2
Lateral	3, 5, 7, 10	1

Direct integration in the physical coordinates is inefficient because the single-particle orbitals occupy only a small part of the nominal domain. We therefore transform every particle to variables matched to the ground-state density. The axial coordinate is first written

$$z = u c \sqrt{1 - r^2/a^2}, \quad 0 < u < 1. \quad (27)$$

The numerical radial domain is restricted to the validated cutoff $R = 0.9a$. Using the ground-state in-plane length ℓ , the reference radial and axial densities are

$$p_r(r) = \frac{1}{C} \frac{2r}{\ell^2} e^{-r^2/\ell^2}, \quad C = 1 - e^{-R^2/\ell^2}, \quad p_u(u) = 2 \sin^2(\pi u). \quad (28)$$

Both reference densities have analytical cumulative distributions. For the radial density, the inverse is

$$r(v) = \ell \sqrt{-\ln(1 - Cv)}, \quad 0 < v < 1, \quad (29)$$

whereas the axial cumulative distribution has no elementary closed form inverse and is tabulated once and interpolated. This transformation cancels most of the localized single-particle density and leaves an order-unity transformed integrand. It is an importance transformation, not a symmetry reduction.

The final estimator removes only the redundant global azimuthal rotation. For the exciton, the hole angle is fixed to zero, the electron angle spans $[0, 2\pi]$, and a factor 2π restores the global rotation. For the biexciton, $\phi_a = 0$ while ϕ_1 , ϕ_2 , and ϕ_b are each integrated independently over $[0, 2\pi]$, again with an overall factor 2π . No reflection folding or particle-exchange multiplicities are used. The complete $G(P + Q)$ factor, the six Coulomb interactions in Eq. (18), and the four kinetic contributions in Eq. (21) remain in the pointwise integrand.

3.2 Adaptive quasi-Monte Carlo quadrature

Every integral is evaluated with the Wolfram Language implementation of adaptive quasi-Monte Carlo quadrature. Low-discrepancy points improve the coverage of the high-dimensional unit hypercube relative to independent pseudorandom sampling [15], while adaptive subdivision directs additional evaluations toward regions with larger estimated error. The production settings are

$$\begin{aligned} \text{Method} &= \text{AdaptiveQuasiMonteCarlo}, \\ \text{BisectionDithering} &= 0, \\ \text{AccuracyGoal} &= \infty, \quad \text{PrecisionGoal} = 4. \end{aligned} \tag{30}$$

The stage-dependent point budget is supplied through `MaxPoints`. A finite absolute accuracy goal is inappropriate because some biexciton numerator and norm integrals are much smaller than a typical absolute tolerance. Reaching `MaxPoints` therefore indicates that the requested relative goal was not attained before the budget cap; the returned integral value and internal error estimate are nevertheless retained.

The Hamiltonian numerator and norm denominator are independent adaptive calls. They do not share a fixed Sobol point set. This preserves separate adaptation to their different integrands. The internal quadrature errors returned for the numerator and denominator are treated as independent and propagated to the exciton and biexciton energies, the binding energy, and the radiative lifetimes. The resulting error bars are numerical-resolution indicators, not statistical confidence intervals: they neglect covariance between the adaptive integrals, possible systematic quasi-Monte Carlo bias, and uncertainty in the location of the variational optimum.

The final estimator was selected through a sequence of checks at $(5, 1)$. These included the exact exciton normalization at $\alpha_X = 0$, integration budget comparisons, dithered adaptive partitions, a complete four-angle biexciton calculation, and the final global-rotation-fixed calculation. The two angular formulations agree within the observed resolution, while removing the redundant global angle reduces the numerical dimension. Earlier direct/cross and particle-relabeling identities remain analytically valid, but their independent numerical evaluations exhibited percent-level differences and were therefore not used to assemble the final energies.

3.3 Variational search

The variational search proceeded in two phases. The first was an exploratory search performed with the Wolfram Language function `NMinimize` using the Nelder–Mead method. It varied the parameters active in each state simultaneously, with a cap of 10^5 points per numerical integral, and located approximate minima for each geometry. This relatively inexpensive phase was used to survey the multidimensional parameter space. Because small objective differences at this budget can be dominated by quasi-Monte Carlo fluctuations, the first-phase optima were used only as starting points.

The second phase was a noise-aware coordinate pattern search with two-budget verification, using the integration setup of section 3.1. A local, derivative-free procedure was chosen because

numerical integration noise makes small objective differences and finite-difference gradients unreliable. The exciton and biexciton searches were implemented differently to reflect their different numbers of variational parameters and evaluation costs.

The exciton objective depends only on α_X , and its kinetic contribution is analytical. The center and neighboring values of α_X were therefore evaluated directly at 2×10^6 points per numerical integral. The best candidate was then tested with a smaller step size on either side, at the same budget, and the lowest result was retained. Screening at a lower budget was unnecessary for this one-dimensional and comparatively inexpensive search.

The biexciton objective depends on $(\alpha, \beta, \gamma, \delta)$, and both its kinetic and potential contributions are evaluated numerically in the full four-particle integral. For this more expensive four-dimensional search, the center and the positive and negative step along each coordinate axis were first screened at 10^6 points per integral. The center and the most promising candidate were then recomputed at 2×10^6 points per integral. A move was accepted only if it lowered the energy at both budgets by more than the observed budget-to-budget drift. Step sizes were progressively reduced around stable accepted candidates, and the same two-budget test was repeated. The lower budget limited the cost of screening the coordinate candidates, while the higher-budget check reduced the chance of accepting a favorable numerical fluctuation.

Automation generated and evaluated the candidates, but final selection also relied on inspecting the energies, integration-error estimates, stability between budgets, neighboring points in parameter space, and consistency of the geometry trends. A parameter set was retained only when this review indicated a stable local minimum rather than an isolated low numerical fluctuation.

4 Results and discussion

4.1 Corrected exciton and biexciton energies

The correlation corrections and corrected energies are collected in [Table 2](#) of the Appendix. The results in [sections 4.1 to 4.3](#) are dimensionless: lengths are scaled by r_B and energies by the effective Rydberg of [Eq. \(2\)](#). At fixed effective-mass ratio $\eta = m_e/m_h$, they are material-scaled and do not require a dimensional GaAs calibration. Conversion to dimensional units is used only for the radiative-lifetime calculation. The values include the single-particle confinement and correlation corrections but exclude the band gap. Because the zeroth-order energies are analytical, each correlation correction has the same numerical integration error as its corresponding corrected energy. Although these rough errors are retained in the table, they are too small to be visible on the scale of the biexciton-energy curves in [Fig. 2](#). The numerical values are reported for transparency and reproducibility, but the physically relevant conclusions are the comparative geometry trends within the idealized model rather than absolute predictions for a particular quantum dot.

Along the axial scan, E_{XX} decreases steeply and nonlinearly, with most of the change occurring between the two most strongly confined geometries. This behavior reflects the strong sensitivity of the confinement energy to the short semiaxis in the oblate regime.

The lateral scan is much flatter: enlarging the in-plane semiaxis produces only a modest reduction in E_{XX} . The axial dimension is therefore the dominant geometrical control of the absolute biexciton energy over the sampled range, consistent with the separation of axial and lateral energy scales in [Eq. \(7\)](#).

4.2 Biexciton binding

The central binding energy is positive throughout the sampled geometries and decreases monotonically when either semiaxis is increased ([Fig. 3](#)). Thus relaxing axial or lateral confinement reduces the net correlation advantage of the four-particle state relative to two isolated excitons. The lateral scan approaches the zero-binding threshold more closely than the axial scan.

The binding is well separated from zero at every geometry except the widest lateral dot. There, the numerical-resolution estimate is comparable to the small central binding energy. Because that

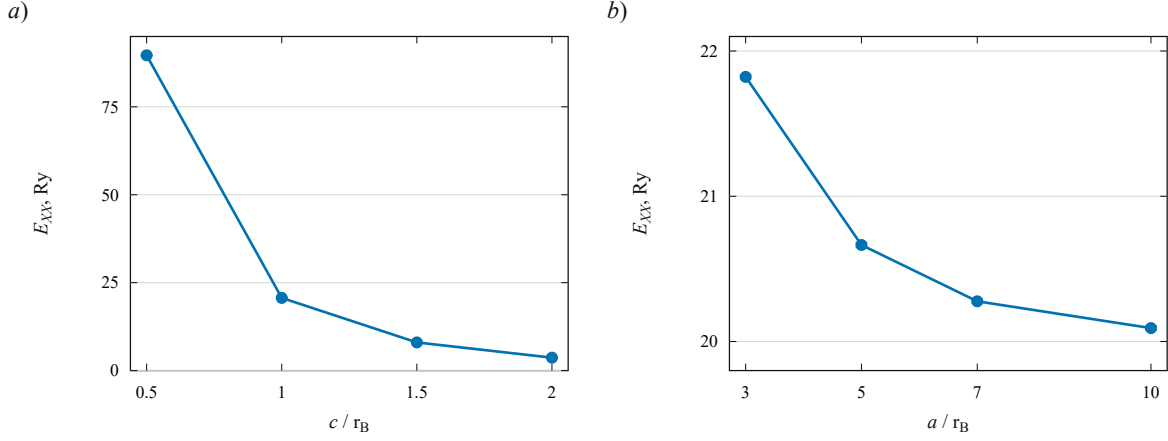


Figure 2: Corrected biexciton energy as a function of (a) the axial semiaxis c at fixed $a = 5r_B$ and (b) the lateral semiaxis a at fixed $c = r_B$, in effective Rydbergs. The band gap is excluded. Lines are guides to the eye, and numerical error bars are smaller than the symbols at the displayed scales.

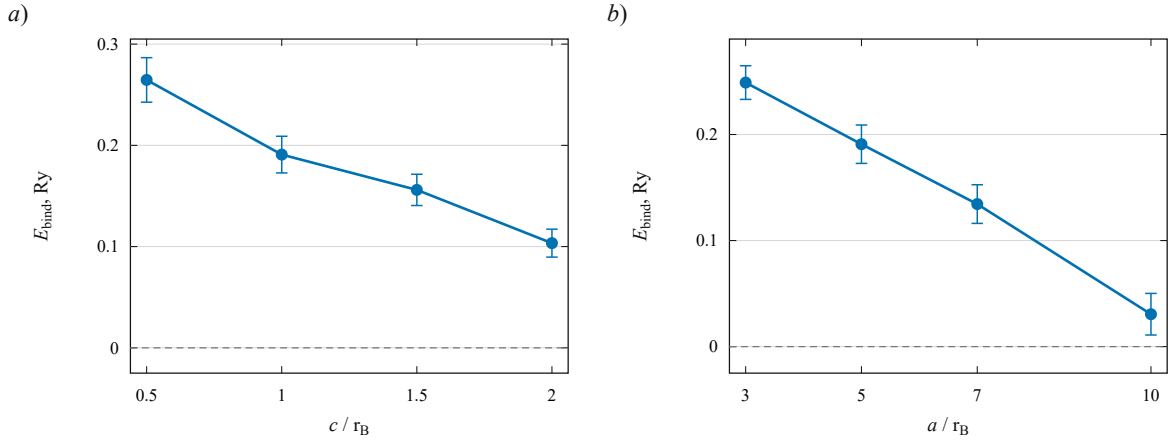


Figure 3: Binding energy $E_{\text{bind}} = 2E_X - E_{XX}$ as a function of (a) c at fixed $a = 5r_B$ and (b) a at fixed $c = r_B$, in effective Rydbergs. Positive values correspond to a bound biexciton. The dashed line marks zero binding. Error bars are rough first-order propagations of the adaptive-integration estimates, not statistical confidence intervals.

estimate omits systematic integration bias and uncertainty in the optimum, we classify the point as marginally positive rather than claiming firmly resolved binding. No sign change is demonstrated within the studied range.

Because the single-particle part cancels in Eq. (23), the origin of this trend is exposed by comparing the two correlation corrections in Fig. 4. Both become less negative as the dot is enlarged, while the separation between $2\Delta E_X$ and ΔE_{XX} narrows. That separation is precisely E_{bind} , so the convergence of the two curves accounts directly for the suppression of binding, particularly along the lateral scan.

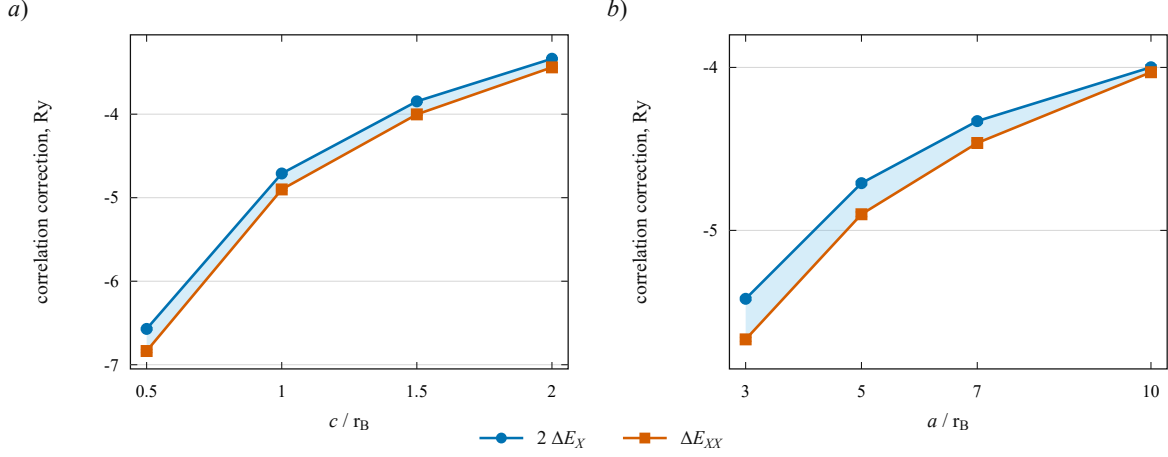


Figure 4: Comparison of the two-exciton correction $2\Delta E_X$ with the biexciton correction ΔE_{XX} as functions of (a) c at fixed $a = 5r_B$ and (b) a at fixed $c = r_B$, in effective Rydbergs. Their vertical separation equals the binding energy. Lines are guides to the eye; numerical error bars are smaller than the plotted symbols at this scale.

4.3 Variational parameters and optical intermediates

The complete optimized biexciton parameters are reported in Table 3 of the Appendix and summarized graphically in Fig. 5. The exciton parameter α_X generally decreases along the axial scan and increases along the lateral scan. For the biexciton, the optimized cross-pair exponent β becomes very small at the larger axial sizes. The composite quantity γ/δ , which sets the maximum of the isolated factor $r_{ab}^\gamma e^{-\delta r_{ab}}$, increases smoothly as either dot dimension is enlarged. This indicates that the hole-hole factor shifts toward larger characteristic separations in larger dots.

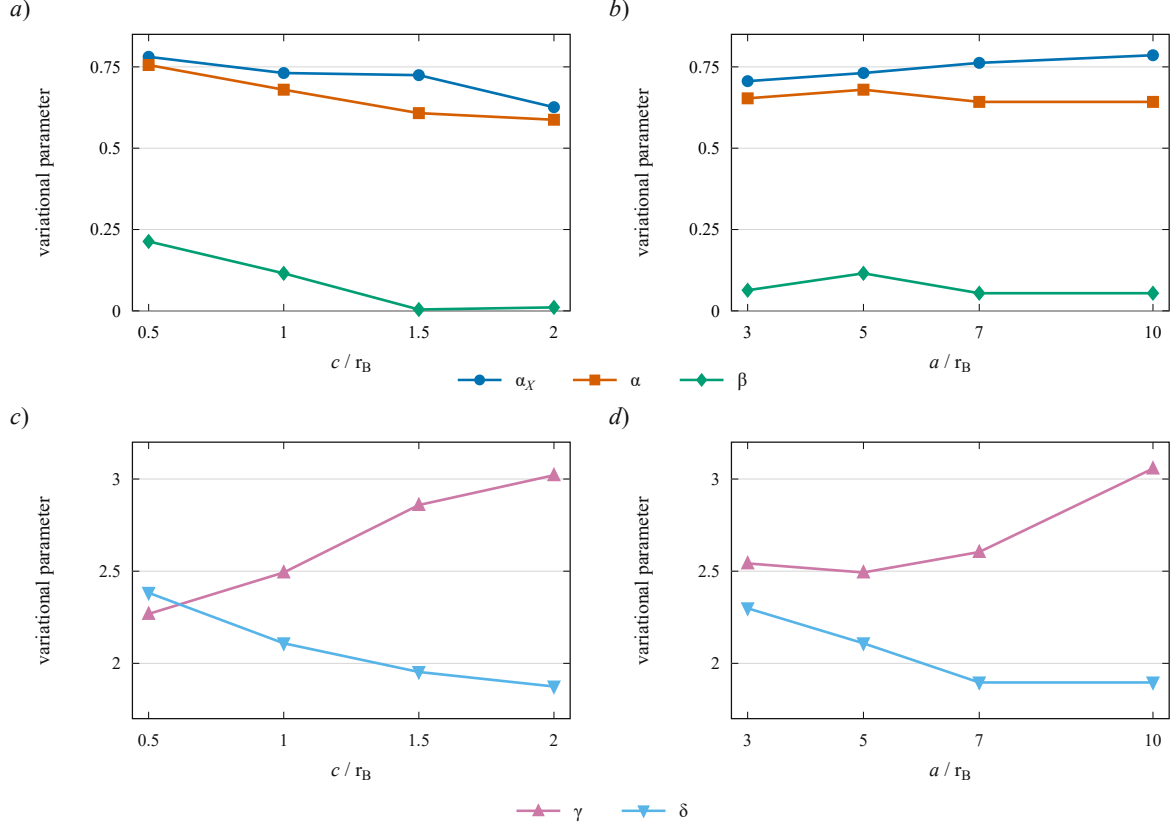


Figure 5: Optimized variational parameters along the axial scan at fixed $a = 5r_B$ and the lateral scan at fixed $c = r_B$. Panels (a) and (b) show α_X , α , and β ; panels (c) and (d) show γ and δ . Lines are guides to the eye.

These parameter-level trends should be interpreted more cautiously than the energies. The objective is numerically rough, the second-phase search is local, and different parameter combinations can produce energies separated by less than the practical quadrature resolution. The parameters characterize the retained variational minimum but are not claimed to be unique.

The coincidence overlap is shown in Fig. 6, while the derived optical quantities are listed in Table 4 of the Appendix. The overlap increases as either dot dimension is enlarged. It tends toward saturation at the larger axial sizes, whereas it continues to rise across the lateral scan. The oscillator strength follows the same behavior.

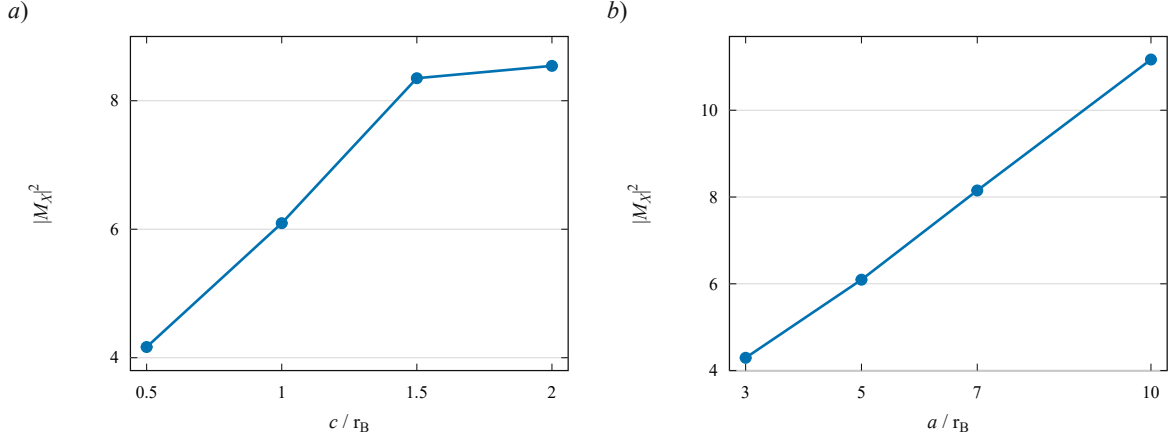


Figure 6: Squared exciton coincidence overlap $|M_X|^2$ as a function of (a) c at fixed $a = 5r_B$ and (b) a at fixed $c = r_B$. Lines are guides to the eye, and numerical error bars are smaller than the plotted symbols.

4.4 Radiative lifetimes

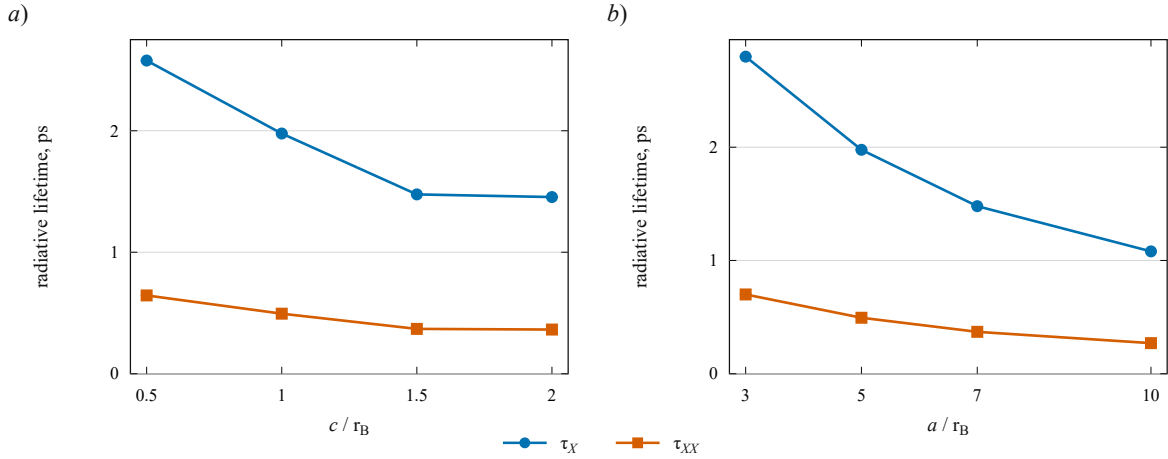


Figure 7: Exciton and biexciton radiative lifetimes as functions of (a) c at fixed $a = 5r_B$ and (b) a at fixed $c = r_B$. The values use the corrected optical transition energy and the coincidence overlap. The adopted decay-channel convention gives $\tau_{XX} = \tau_X/4$. The single GaAs parameter set is specified in [section 2.4](#). Propagated error bars are smaller than the plotted symbols.

Both radiative lifetimes decrease as the dot is enlarged ([Fig. 7](#)). Along the axial scan, the decrease weakens and becomes nearly saturated for the largest dots. Along the lateral scan, the decline remains pronounced across the sampled range. The biexciton lifetime follows the same geometry dependence at one quarter of the exciton value.

Relaxing confinement decreases E_{exc} , which alone would tend to increase the lifetime. The simultaneous increase in $|M_X|^2$ is stronger, however. Since substitution of [Eq. \(25\)](#) into [Eq. \(24\)](#) gives $\tau_X \propto (E_{\text{exc}}|M_X|^2)^{-1}$, the overlap enhancement dominates and the lifetime decreases.

4.5 Comparison and limitations

The present geometry trends are qualitatively consistent with calculations for full strongly oblate ellipsoidal dots, where the semiaxes tune the correlated energies and binding and the exciton overlap sets the radiative scale [[9](#), [10](#)]. A direct numerical comparison is not appropriate, however,

because the semi-ellipsoidal boundary changes the axial single-particle state, and the present four-particle trial function and full integration protocol differ from those earlier calculations.

The model assumes impenetrable barriers, bulk effective masses, static dielectric screening for the energy calculation, and an adiabatic single-particle ground state. Finite band offsets, dielectric mismatch, strain, atomistic symmetry, and phonon-assisted decay are outside the present scope. The lifetime calculation follows the optical model and decay-channel convention of Ref. [9], using the single GaAs parameter set specified in [section 2.4](#). It should therefore be interpreted as the radiative lifetime within that model rather than as a complete experimental decay time.

These approximations can shift the absolute energies, binding energies, and lifetimes of a real structure. The principal outcome is therefore the qualitative separation of axial and lateral geometry trends within a single consistent model, not a quantitative prediction for a specific fabricated dot.

Numerically, the reported bars quantify only first-order propagation of the adaptive integrator’s internal errors. The smooth trends and the positive binding away from the widest lateral dot are robust on that scale, but a higher-resolution targeted calculation would be required to determine firmly whether the marginal endpoint is bound or antibound.

5 Conclusions

We have calculated the ground exciton and biexciton states of a strongly oblate semi-ellipsoidal GaAs quantum dot over intersecting axial and lateral geometry scans. A ground-orbital-matched importance transformation, a full global-rotation-fixed biexciton estimator, and the second-phase search with two-budget verification were combined to control the numerical sensitivity of the small binding energy.

Within the hard-wall effective-mass model, the main conclusions are qualitative. The axial dimension controls the absolute confinement-energy scale much more strongly than the lateral dimension. Enlarging the dot along either direction weakens the biexciton binding, with the lateral scan approaching the zero-binding threshold most closely. The central binding estimate remains positive across the sampled range, but its sign is not resolved at the widest lateral geometry.

Increasing either semiaxis also enhances the exciton coincidence overlap and oscillator strength. This enhancement outweighs the reduction of optical transition energy and shortens both calculated radiative lifetimes. The lateral dependence remains pronounced across the sampled range, whereas the axial dependence tends toward saturation.

The numerical values should therefore be interpreted as results of a controlled model comparison rather than as absolute predictions for a particular fabricated dot. Their principal contribution is to identify distinct geometrical controls: the short axial dimension sets the dominant confinement-energy scale, while lateral enlargement provides an effective route for suppressing weak biexciton binding and enhancing radiative strength. Extensions to finite barriers, dielectric mismatch, strain, and atomistic confinement are needed for material-specific quantitative predictions. A targeted higher-resolution calculation is also required to determine the sign of the near-zero binding at the widest lateral geometry.

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A Supplementary numerical information

This appendix collects the numerical tables supporting the main-text figures. Table 2 gives the correlation corrections, corrected energies, and binding energies.

Table 2: Exciton and biexciton correlation corrections, corrected energies, and binding energy, all in effective Rydbergs. Errors are rough propagated numerical integration estimates.

a	c	ΔE_X	E_X	ΔE_{XX}	E_{XX}	E_{bind}
5	0.5	-3.286 ± 0.007	44.958 ± 0.007	-6.836 ± 0.018	89.652 ± 0.018	0.265 ± 0.022
5	1	-2.355 ± 0.006	10.428 ± 0.006	-4.901 ± 0.014	20.665 ± 0.014	0.191 ± 0.018
5	1.5	-1.923 ± 0.005	4.079 ± 0.005	-4.002 ± 0.011	8.002 ± 0.011	0.156 ± 0.015
5	2	-1.668 ± 0.005	1.889 ± 0.005	-3.440 ± 0.010	3.674 ± 0.010	0.103 ± 0.014
3	1	-2.710 ± 0.005	11.035 ± 0.005	-5.669 ± 0.012	21.822 ± 0.012	0.249 ± 0.016
7	1	-2.165 ± 0.006	10.206 ± 0.006	-4.464 ± 0.013	20.277 ± 0.013	0.134 ± 0.018
10	1	-1.999 ± 0.007	10.061 ± 0.007	-4.029 ± 0.014	20.092 ± 0.014	0.031 ± 0.020

Table 3 lists the final parameters retained by the second-phase search. The ratio γ/δ is also shown because it locates the maximum of the isolated hole-hole factor $r_{ab}^\gamma e^{-\delta r_{ab}}$.

Table 3: Final optimized parameters for the exciton and biexciton states.

a	c	α_X	α	β	γ	δ	γ/δ
5	0.5	0.780939	0.755694	0.213328	2.268187	2.382724	0.952
5	1	0.731089	0.679864	0.115335	2.492963	2.108847	1.182
5	1.5	0.724547	0.607794	0.004089	2.858881	1.953280	1.464
5	2	0.626090	0.587481	0.010584	3.020600	1.874374	1.612
3	1	0.706089	0.653302	0.063307	2.542182	2.298690	1.106
7	1	0.762339	0.642364	0.054180	2.604026	1.896347	1.373
10	1	0.785787	0.642364	0.054180	3.056900	1.896347	1.612

The coincidence overlap and the optical quantities derived from it are listed in Table 4.

Table 4: Coincidence overlap, oscillator strength, and radiative lifetimes using the GaAs parameter set of section 2.4. Numerical-resolution estimates are omitted because they are small compared with the geometry-dependent variations.

a	c	$ M_X ^2$	f_X	τ_X (ps)	τ_{XX} (ps)
5	0.5	4.167	53.48	2.578	0.644
5	1	6.094	87.75	1.977	0.494
5	1.5	8.351	122.99	1.476	0.369
5	2	8.543	126.82	1.454	0.364
3	1	4.294	61.70	2.800	0.700
7	1	8.151	117.45	1.479	0.370
10	1	11.169	161.02	1.080	0.270